Yung-Chin (Jim) Chen

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Research Interests

In-Memory Computing Systems, LLM Quantization, Efficient LLM Inference, VLSI Design

Education

Princeton University — Princeton, NJ, USA
Ph.D. Candidate in Electrical and Computer Engineering — *Sep 2024 – Present*

National Taiwan University (NTU) — Taipei, Taiwan
Bachelor of Science in Electrical Engineering — *Sep 2019 – Jan 2024*

- Phi Tau Phi Honorary Member (for top 1% of college graduates)

Keio University — Tokyo, Japan
Exchange Student — *Oct 2022 – Jul 2023*

Research Experience

Ph.D. Student at Verma Lab — Jan 2025 – Present
Princeton University (Advisor: Prof. Naveen Verma) — *Princeton, NJ, USA*

- Researching LLM quantization for in-memory computing (IMC) systems and architectures
- Researched the mechanistic interpretability of massive activations in LLMs

Visiting RA at Computing and Sensing Group — Sep 2022 – Feb 2024
Keio University (Advisor: Prof. Kentaro Yoshioka) — *Kanagawa, Japan*

- Researched saliency-aware IMC macros for neural network inference
- Researched memory-centric algorithm-architecture co-design methods for neural network frameworks

Undergrad RA at Energy-Efficient Circuits and Systems Lab — Sep 2021 – Feb 2024
National Taiwan University (Advisor: Prof. Tsung-Te Liu) — *Taipei, Taiwan*

- Taped out a 28nm SRAM IMC-based accelerator for end-to-end neural network inference

Publications

[1] **Yung-Chin Chen**, Ourania Stouraiti, Naveen Verma, "Hessian-Guided Mixed ADC Precision for Analog In-Memory Computing Systems on LLM Inference", *Under Review*

[2] **Yung-Chin Chen**, Chung Peng Lee, Ze-Wei Liou, Naveen Verma, "Massive Spikes in LLMs are Bias Vectors: Mechanistic Uncovering and Spike-Free Quantization", *Under Review*

[3] Wenlun Zhang, Shimpei Ando, **Yung-Chin Chen**, Kentaro Yoshioka, "ASiM: Modeling and Analyzing Inference Accuracy of SRAM-Based Analog CiM Circuits", *in IEEE Transactions on Very Large Scale Integration Systems (TVLSI), 2025*

[4] Shimpei Ando, Satomi Miyagi, Wenlun Zhang, **Yung-Chin Chen**, Kentaro Yoshioka, "A 4541 TOPS/W Saliency-Aware Analog Computing In-Memory Macro with Charge-Domain Saliency Detector", *in IEEE European Solid-State Electronics Research Conference (ESSERC), 2025*

[5] **Yung-Chin Chen**, Shimpei Ando, Daichi Fujiki, Shinya Takamaeda-Yamazaki, Kentaro Yoshioka, "OSA-HCIM: On-The-Fly Saliency-Aware Hybrid SRAM CIM with Dynamic Precision Configuration", *in Asia and South Pacific Design Automation Conference (ASP-DAC), 2024*

[6] Wenlun Zhang, Shimpei Ando, **Yung-Chin Chen**, Satomi Miyagi, Shinya Takamaeda-Yamazaki, Kentaro Yoshioka, "PaCiM: A Sparsity-Centric Hybrid Compute-in-Memory Architecture via Probabilistic Approximation", *in International Conference on Computer-Aided Design (ICCAD), 2024*

[7] **Yung-Chin Chen**, Shimpei Ando, Daichi Fujiki, Shinya Takamaeda-Yamazaki, Kentaro Yoshioka, "HALO-CAT: A Hidden Network Processor with Activation-Localized CIM Architecture and Layer-Penetrative Tiling", *on arXiv*

[8] Kentaro Yoshioka, Shimpei Ando, Satomi Miyagi, **Yung-Chin Chen**, Wenlun Zhang, ”Towards Efficient and Precise Analog Compute-in-Memory Circuits”, *In International Conference on Solid State Devices and Materials (SSDM), 2024 (Invited)*

[9] Kentaro Yoshioka, Shimpei Ando, Satomi Miyagi, **Yung-Chin Chen**, Wenlun Zhang, ”A Review of SRAM-based Compute-in-Memory Circuits”, *in Japanese Journal of Applied Physics (JJAP), 2024*

[10] Shimpei Ando, **Yung-Chin Chen**, Satomi Miyagi, Wenlun Zhang, Kentaro Yoshioka, ”A Saliency-Aware Analog Computing In-Memory with SAR-Embedded Detection Achieving 18.5% Power Reduction”, *in Japanese Journal of Applied Physics (JJAP), 2024*

Relevant Coursework

Princeton University
Computer Architecture · Domain-Specific Computer Systems and Architectures · Design of VLSI Systems · Efficient Systems for Foundation Models · Machine Learning and Pattern Recognition · Introduction to Reinforcement Learning

National Taiwan University
Integrated Circuit Design · Integrated Circuit Design Lab · Introduction to Electronic Design Automation · Digital Circuit Lab · Digital Signal Processing in VLSI Design · Computer Architectures · Switching Circuit and Logic Design · Electronic Circuits · Microelectronics I & II · Machine Learning · Signals and Systems · Probability and Statistics

Honors and Awards

Study Abroad Scholarship - Japan Student Services Organization	Sep 2022 – Jul 2023
Irving T. Ho Memorial Scholarship - EE dept. at NTU	Dec 2022
The Memorial Scholarship Foundation to Lin Hsiung Chen (acc. rate: 2%)	Nov 2022
Research Grant - National Science and Technology Council (NSTC), Taiwan	Jul 2022 – Jan 2023
Research Grant - Taiwan Semiconductor Manufacturing Co., Ltd (TSMC)	Feb 2022 – Jun 2022
1st Place (Best Solver Award) - MakeNTU - largest student maker hackathon in Taiwan	Mar 2022
Principal Award - Awarded to top 2% of NTUEE students for academic excellence	Dec 2022

Teaching Experience

TA for Design of VLSI Systems, taught by Prof. Naveen Verma	Sep 2025 – Dec 2026
TA for Computer Architecture, taught by Prof. Tsung-Te Liu	Sep 2023 – Jan 2024
TA for EECS Lab Undergraduate Research, supervised by Prof. Tsung-Te Liu	Sep 2023 – Jan 2024
TA for Signals and Systems, taught by Prof. Lin-Shan Lee	Feb 2022 – Jun 2022

Technical Skills

Programming: Verilog, System Verilog, Python, C++, MATLAB
IC Design Tools: NC-Verilog, Design Compiler, Innovus, Virtuoso, HSPICE, FPGA
Tools and Frameworks: PyTorch, TensorFlow, NumPy, LaTeX
Languages: Chinese (Native), English, Japanese

Activities

Vice President of Princeton Association of Taiwanese Students	Sep 2025 – Present
Compulsory Military Service in Taiwan	Mar 2024 – Jun 2024
Student Ambassador of National Taiwan University	Oct 2021 – Sep 2022
Vice Captain of NTUEE Baseball Team	Sep 2021 – Aug 2022
Vice President of NTU Escape Room and Puzzle Solving Club	Aug 2020 – Jul 2021